Is Mining Worth It? Local Effects of Legal Mineral Production in Brazil*

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Abstract

This paper examines the relationship between legal mineral production and local economic development in Brazil. We compare municipalities where prospecting licenses resulted in economically viable deposits with municipalities whose prospecting never led to commercial production, an outcome close to random conditional on having applied. Using a staggered difference-in-differences design across iron, gold, and bauxite, we find that producing iron and bauxite municipalities settle onto a permanently higher level of local GDP per capita. For iron, the gain is accompanied by modestly higher wages, more manufacturing employment, and less deforestation, with no change in overall employment or crime. For bauxite, only the level of GDP per capita moves; employment, wages, crime, and deforestation are all null. Gold is mostly null: no effect on GDP, wages, crime, or deforestation, and only a fragile gain in the employment rate, consistent with a large informal artisanal gold sector. Legal mining in Brazil makes municipalities richer without making them more dangerous or more deforested.

Keywords: Mining; Local Development; Difference-in-Differences.
JEL Codes: J3, O13, O15, Q33.

1 Introduction

Mineral resources are frequently promoted as engines of local development: royalty payments, employment, and broader economic spillovers are the standard justification for granting exploration and extraction rights in a given municipality. Whether this promise is kept is an empirical question with a long and contested history. Answering it is especially

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important for the Brazilian case, given the importance of the mineral sector: mineral exports averaged approximately 15% of the trade balance between 2010 and 2023, reaching peaks above 20% in 2011 and 2021, while mineral production corresponded to 2.1% of GDP on average over the same period.[1]

This paper asks whether hosting mineral production brings local development gains to the municipalities involved. We study three of Brazil's four most economically significant solid minerals in terms of Financial Compensation for the Exploitation of Mineral Resources (CFEM) revenue: iron, gold, and bauxite.[2] We trace the consequences of mining across four outcome families: economic growth (GDP per capita and its growth rate), local labor markets (employment, employment rates, manufacturing employment, and wages), violence (homicide rates), and the environment (deforestation). To do this, we exploit a quasi-experimental design based on Brazil's mineral prospecting regime, treating each mineral as a separate lottery to ask not only whether mining generates net local benefits, but whether those effects depend on what is being extracted.

This study is part of a broader debate on whether natural resource abundance helps or hinders economic development. Cross-country evidence on the "natural resource curse" finds that resource-abundant countries often grow more slowly and develop weaker institutions than resource-poor ones (Badeeb et al., 2017; Havranek et al., 2016; Stijns, 2005; Sachs & Warner, 2001), while other work finds that resource wealth is compatible with, and sometimes a driver of, sustained growth (Larsen, 2005; Auty, 2001).

However, much of the cross-country evidence faces important empirical challenges (Van Der Ploeg & Poelhekke, 2019). Resource dependence variables are often endogenous to the very development outcomes they aim to explain, macroeconomic data are frequently limited in quality, and cross-country regressions remain vulnerable to omitted variable bias and a wide range of confounding factors. These limitations make it difficult to distinguish the effects of natural resource abundance itself from those of other characteristics associated with resource-rich countries, potentially leading to incorrect attribution of poor development outcomes to resource abundance.

We follow a recent strand of the literature that exploits within-country variation in resource discoveries to estimate the local effects of mining (Katovich, 2024). Within-country designs provide a valuable alternative because they allow researchers to control for common institutional and structural factors that also influence economic performance, thereby improving the identification of causal effects.

At the local level, the mechanisms through which mining can fail to deliver broad-based development are distinct from the aggregate channels emphasized in the cross-country literature (Allcott & Keniston, 2018; Jacobsen & Parker, 2016). A mine may draw local labor and capital away from more dynamic activities. It may operate as an enclave with few linkages to the surrounding economy, generating output that does not translate into

[1] Sources: MDIC/Comex Stat, https://comexstat.mdic.gov.br/pt/geral, and Anuário Mineral Brasileiro - ANM, https://dados.gov.br/dados/conjuntos-dados/anuario-mineral-brasileiro-amb.

[2] The fourth, copper, is excluded from our analysis because too few municipalities meet our treatment definition (Section 2.1) to support a separate lottery.

local employment and sustained growth. Royalty payments may produce fiscal windfalls without generating productive public investment. Within-country designs that exploit plausibly exogenous variation in resource discovery are better suited to identify these local mechanisms. Caselli & Michaels (2013) use Brazilian oil discoveries to show that oil windfalls raise public revenue and spending without commensurate gains in household living standards. Cavalcanti et al. (2019), whose research-authorization lottery design we adapt from oil to mining, find that discovery municipalities experience a lasting increase in output but with limited effects on local labor markets.

Resource booms may carry social costs that do not show up in GDP. Rapid in-migration and transient populations with weak community ties, coupled with local institutions that struggle to keep pace with economic change, have been linked to elevated crime in extraction settings (Ruddell, 2011; Carrington et al., 2010; Broadway, 2000). The capture of mining rents by criminal organizations is itself a documented driver of violence (Herrera & Martinez-Alvarez, 2022; Berman et al., 2017). On the environmental margin, illegal and informal mining in the Brazilian Amazon has been linked directly to deforestation (Siqueira-Gay & Sánchez, 2021; Sonter et al., 2017). This literature motivates our focus on growth, local labor markets, homicides, and deforestation, while raising a question about whether the social costs documented for illegal mining extend to the legal, royalty-paying sector.

Our lottery design takes advantage of the regulatory framework governing mineral exploration in Brazil, which generates a quasi-experiment from the plausibly random outcome of prospecting. Before any mineral extraction can begin in Brazil, a firm must request a research (prospecting) authorization from the National Mining Agency (ANM) and carry out technical studies assessing whether the deposit is economically viable. Authorizations are granted in many locations with similar observable geological and geographic characteristics, but only a subset of them result in a viable discovery. Conditional on having applied for a research authorization, whether a given area contains an economically exploitable deposit is plausibly close to random with respect to local socioeconomic characteristics. Our identification strategy therefore compares municipalities where prospecting yielded an economically viable deposit against municipalities that prospected for the same mineral but never produced, whether because no viable deposit was found or because a title was issued but never translated into commercial extraction.

Our central finding is that legal mineral production delivers real but uneven local benefits. Iron municipalities settle onto a permanently higher GDP per capita, roughly 25%, with accompanying gains in wages and manufacturing employment and less deforestation, but no change in overall employment or crime. Bauxite municipalities also see a permanently higher GDP per capita, roughly 44%, but with no detectable effect on employment, wages, deforestation, or crime. In both minerals, post-treatment GDP growth rates are flat or marginally negative, consistent with a one-time shift in the level of activity rather than a permanently faster growth path. Gold is mostly null: we find no detectable effect on GDP, wages, crime, or deforestation, and only a fragile gain in the employment rate. This pattern is consistent with a large informal artisanal sector, which is poorly captured by the formal royalty record.

We contribute to the within-country resource literature by extending the discovery-lottery design from oil (Cavalcanti et al., 2019) to solid minerals, showing that "mining" is not a single treatment. Iron, bauxite, and gold leave very different local footprints, possibly reflecting differences in how each mineral is extracted and processed. A single-commodity design could not have recovered this result. This contribution carries a policy implication: legal mining is locally beneficial but narrowly so. Output and royalties reach the municipal accounts, but broad-based employment gains do not follow, not even in iron, where the labor-market response is limited to a small manufacturing nucleus.So the first-order policy question is not whether to permit mining, but how to make its fiscal windfall permeate the local labor market and foster the development of other economic activities that can sustain growth after mineral resources are depleted.

2 The Mineral Exploration Lottery

We construct a separate sample for each mineral – iron, gold, and bauxite. Each sample draws from the universe of municipalities where ANM granted a research authorization and prospecting was actually conducted, defined as an authorization event followed by a communicated start of research or a filed research report in the Mining Registry (ANM, 2024*b*). Within this universe, whether prospecting culminates in commercial production is the quasi-random event our design exploits.

2.1 Treatment Definition

We define treatment at the municipality-mineral level using CFEM, Brazil's mineral royalty. A municipality is treated if its cumulative CFEM payments for that mineral reach at least R$100,000 over the royalty system's full history. We impose this floor because the royalty record contains municipality-mineral pairs with positive CFEM whose implied production is economically insignificant – trace payments that do not correspond to meaningful mineral extraction. These sub-floor pairs are excluded from the sample entirely: 89 for gold, 29 for iron, and 13 for bauxite.

Control municipalities fall into two groups: (a) those that prospected but never obtained a viable deposit or an active title ("prospected, no discovery"), and (b) those that hold an active title and file Annual Mining Reports (RAL) but have never paid CFEM ("licensed, never produced").[3] Treatment status does not require a mining title: 12 iron and 5 gold municipalities pay CFEM without ever appearing in the RAL registry, typically operations licensed under regimes outside the standard authorization–concession flow (*garimpo* permits and extraction licenses).[4] What determines treatment is documented

[3] The *Relatório Anual de Lavra* (RAL, Annual Mining Report) is the activity report that every holder of a mining title must file with ANM each year, covering the previous calendar year (ANM, 2024*b*). Filing is mandatory for as long as the title is active, whether or not the titleholder extracted anything, so the RAL documents the existence and timing of a title but not, by itself, whether commercial production took place.

[4] *Garimpo* – artisanal and small-scale mining – is not synonymous with illegal mining, although the two

commercial production, however licensed.

Municipalities whose cumulative CFEM reaches the threshold for more than one mineral are excluded entirely from the lottery sample. This avoids cross-mineral contamination: a municipality hosting both an iron and a bauxite operation, for example, would be classified as treated in both lotteries. In such cases, it would not be possible to separately identify the causal effect associated with each mineral, potentially biasing the estimation. The screen counts above-floor CFEM in any of the minerals.

The three groups that make up each lottery are therefore the following. *Treated* stands for municipality-mineral pairs whose cumulative CFEM for the mineral reaches the floor. *Control* stands for municipalities that prospected without discovery, or that hold an active title but never paid CFEM. *Excluded* stands for municipalities with positive but sub-floor cumulative CFEM (below floor), or with CFEM at or above the floor for more than one mineral (multi-mineral); both groups are removed from the estimation sample. Sample counts by group are reported in Table 1, and Figure 1 maps the three groups for each lottery.

Table 1: Composition of the Three Mineral Lotteries (CFEM $\geq$ R\$100k Treatment Definition)

Mineral	Treated	Control: Prospected	Control: Licensed, No CFEM	Below Floor	Multi-mineral
Iron	56	1,245	90	29	14
Gold	70	1,850	244	89	15
Bauxite	22	220	47	13	5

Source: ANM (2024*b*) and ANM (2024*a*). Counts are municipality-mineral pairs. The two "Control" columns split the control group defined in the text; "Below floor" and "Multi-mineral" are the two excluded groups. All group definitions are given in Section 2.1.

2.2 *Treatment Timing*

CFEM only began operating as a royalty system in 2002/03, so our data cover royalty payments from that year onward. Dating treatment by the first royalty payment alone would classify all mines that were already producing before 2002/03 as treated only from the first year of CFEM collection, including some mines that had been active since the 1940s (Figure 2 shows how far back each lottery's prospecting history runs). Our main specification therefore dates treatment as $G_i = \min(\mathit{first\_ral}_i, \mathit{first\_cfem}_i)$: CFEM decides

are often conflated. The distinction runs along two different dimensions. Garimpo describes *how* production is organized: low capital intensity, little mechanization, and low labor formalization, in contrast with capital- and technology-intensive industrial mining. Legality describes whether an operation, garimpo or industrial, complies with environmental licensing and ANM rules: garimpo is legal when it operates within them, and illegal when it operates outside them. In gold, informality and illegality are particularly widespread, partly because gold's high value per unit of weight eases transport, sale, and evasion of state controls (Hook, 2019); gold and diamonds are recognized worldwide as especially prone to illegality (Berman et al., 2017).

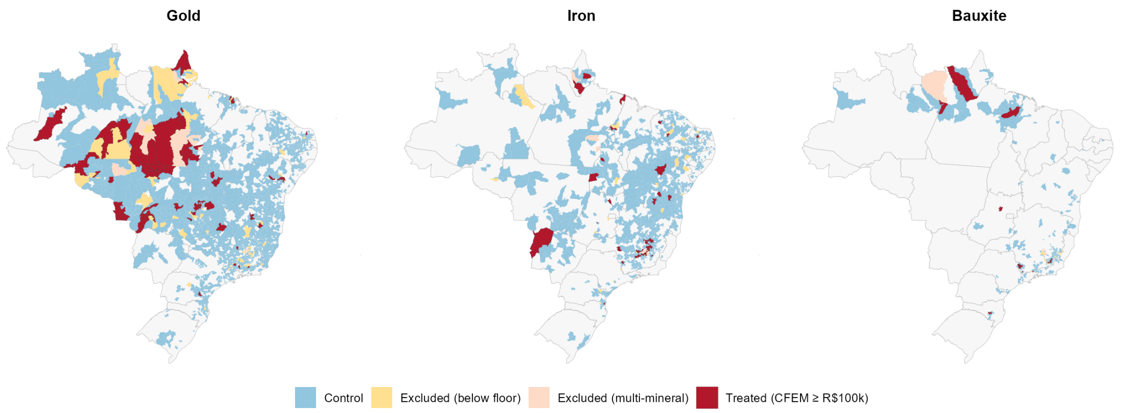

Figure 1: The Three Mineral Lotteries: Treated, Control, and Excluded Municipalities

Source: Own elaboration, based on ANM (2024*b*) and ANM (2024*a*). Group definitions are given in Section 2.1.

who is treated, and the first Annual Mining Report (RAL), filed by every titleholder since as early as 1945, decides *when*, wherever it exists. For the treated municipalities with no mining registry record, the first royalty payment year is the only available date.

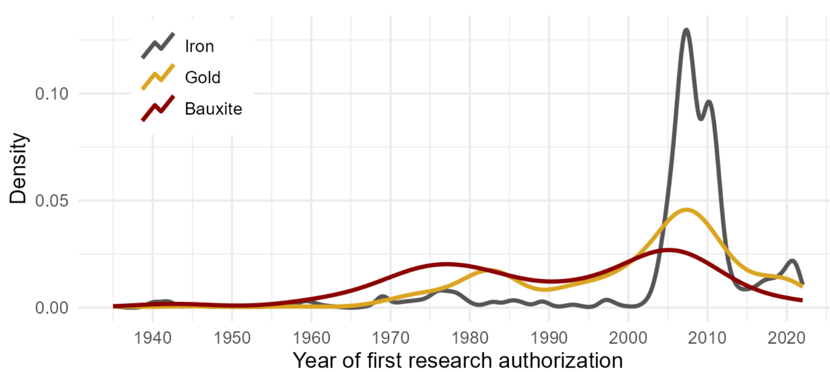

Figure 2: When Prospecting Began: First Research Authorization Years by Mineral

Source: Own elaboration, based on ANM (2024*b*). Kernel densities of the year of the first research authorization across each mineral's lottery pool (treated and control municipalities). Prospecting spans eight decades, with waves in the 1970s–80s and after 2000; this is why treatment cohorts range from the 1940s to the 2020s and why timing and left-censoring require the care described in the text.

We treat the RAL-timed specification as our primary specification and the specification based on the first CFEM payment as a robustness check. The latter defines treatment as any positive CFEM payment by a titled municipality, with no minimum floor, and dates treatment solely by the first payment.

The estimator cannot recover effects for cohorts whose production began before the first year for which data on the outcome are available, as no untreated period is observed for these cohorts. This issue arises for the employment and deforestation outcomes, whose underlying databases begin in 2000, and for the homicide outcome, whose underlying database begins in 2001.

3 Data and Empirical Strategy

3.1 Outcome Variables and Data Sources

We estimate effects on eight outcomes across four families. *Labor market*: log employment rate (formal employment over total municipal population), log employment level, log manufacturing employment, and log average wage in minimum wages (RAIS, 2024). *Local economic activity*: log GDP per capita and log-difference GDP growth rate (IPEA municipal GDP series). *Crime*: log homicide rate per 100,000 population (Atlas da Violência). *Environment*: log deforestation flow, defined as the log of the annual increment in cumulative deforested area plus one (PRODES/INPE). Municipal units are held fixed over time using geobr's Minimum Comparable Areas (AMCs)(Pereira & Barbosa, 2026). In all regressions, labor market, economic activity, and crime outcomes are estimated giving greater weight to larger municipalities (population weights, IBGE); the deforestation outcome gives greater weight to municipalities with larger land area (area weights, IBGE).

3.2 Econometric Specification

We estimate the group-time average treatment effects of Callaway & Sant'Anna (2021), applied separately to each mineral's lottery. Let $Y_{m,t}$ denote an outcome for municipality (AMC) m in year t, and let G_m denote the year in which commercial production of the mineral begins in m (Section 2.2); for control municipalities, production never starts and $G_m = \infty$. For each production cohort g, we estimate

$$ATT(g,t) = \mathbb{E}\left[Y_{m,t}(g) - Y_{m,t}(\infty) \,\middle|\, G_m = g\right], \quad t \geq g, \tag{1}$$

the average effect of mineral production on the municipalities whose mines entered production in year g, evaluated in year t. The counterfactual $Y_{m,t}(\infty)$ – the path these municipalities would have followed had prospecting never resulted in production – is recovered from the lottery's losers: municipalities in the same mineral's pool with no commercial production by year t. Identification thus rests on the lottery logic of Section 2: conditional on having prospected, municipalities where production never materialized trace the counterfactual path of those where it did. We estimate $ATT(g,t)$ for every (g,t) using a doubly robust estimator, and aggregate these group-time estimates in two ways. The *simple* aggregation weights each $ATT(g,t)$ by the relative size of cohort g among all treated observations, so that cohorts observed for more post-treatment periods contribute more to the overall average. The *group* aggregation instead first averages $ATT(g,t)$ within each cohort g across its own post-treatment periods, and then averages across cohorts weighting each cohort equally by its size, regardless of how many post-treatment periods it has been observed for.

We base all inference on the *group* aggregation. Because our cohorts span from the 1940s to the 2020s, the simple aggregation would let cohorts treated decades ago dominate the reported average purely because they contribute more calendar-year observations, not because their treatment effect is more representative (Roth et al., 2023; Goodman-Bacon, 2021). We report the simple aggregation alongside the group aggregation in every table for transparency.

4 Results

We present results mineral by mineral – iron, bauxite, then gold. We base inference on the group aggregation and report the simple aggregation alongside it for transparency. We screen every estimate against a pre-treatment event-study check, dropping from the tables and from the paper's conclusions any coefficient with a significant pre-trend.

4.1 Iron

Table 2 reports outcomes for iron. The headline finding is a persistently higher level of local GDP per capita. The coefficient of 0.227 log points corresponds to an increase of approximately 25 percent[5].

Part of this gain reaches workers. The wages rise by about 6 percent, while manufacturing employment per capita rises sharply (+0.572), consistent with a small industrial nucleus forming around the mine rather than broad-based labor market gains. Its magnitude (≈77% in per-capita terms) seems very large as an average economic response and likely reflects small denominators: where the manufacturing base is near zero, a mine's arrival can multiply it severalfold. The estimate itself, however, does not hinge on any single municipality (Section 4.3).

Counterintuitively, deforestation falls (−0.202). One possible explanation is that legal mining brings monitoring and enforcement to the area, displacing the illegal logging that would otherwise take place there. The remaining outcomes – homicides, the employment rate, total employment, and the GDP growth rate – are all null.

4.2 Bauxite

For bauxite, the coefficient of 0.366 log points corresponds to an increase in GDP per capita of approximately 44 percent. Every other outcome – homicides, employment (rate

[5] No single municipality drives this result; the coefficient is stable across all leave-one-out samples (Section 4.3).

Table 2: Iron: Local Effects

Outcome	Simple	Group	Obs.
Homicides (log rate/100k)	−0.219	0.116	42,171
	(0.884)	(0.561)	
Employment rate (log)	0.079	0.042	42,040
	(0.131)	(0.071)	
Employment (log)	0.176	0.065	42,040
	(0.193)	(0.049)	
Wage (log, minimum wages)	0.030	0.060***	42,040
	(0.065)	(0.016)	
Manufacturing employment (log)	0.420*	0.572***	16,199
	(0.232)	(0.092)	
GDP per capita (log, pop-wt)	**0.273****	**0.227*****	**28,423**
	(0.107)	(0.082)	
GDP growth (log-diff, pop-wt)	0.011	0.040	27,049
	(0.037)	(0.032)	
Deforestation (log flow, PRODES)	−0.297*	−0.202***	31,142
	(0.180)	(0.052)	

Group-time ATT from Callaway & Sant'Anna (2021); inference is based on the group column. Treatment: cumulative CFEM $\geq$ R\$100k (Section 2.1); $G_i = \min(\mathit{first\_ral}_i, \mathit{first\_cfem}_i)$ (`cs_cfem_floor100k.Rda`). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Bold row is the paper's headline finding. Manufacturing employment passes the pre-trend check and is reported. "Obs." counts municipality-year observations in the estimation sample; it varies across rows because the outcome panels differ in coverage (Section 3) and because cohorts whose production began before an outcome panel's first year have no untreated period and are dropped by the estimator (Section 2.2).

and level), wages, and deforestation – is null under the group aggregation. The GDP growth rate is marginally negative (−0.089). This pattern, a positive level effect paired with no positive growth rate, is consistent with a one-time level shift followed by ordinary growth. However, the growth-rate coefficient's own pre-treatment path is noisy, so we do not interpret it.

We report the bauxite GDP effect despite a pre-trend caveat. The event-study path shows pointwise-significant deviations at $t = -4$ and $t = -2$, of opposite signs (see Figure A1, panel (b), Appendix). However, a leave-one-out check already addresses the sharpest version of this concern: the estimate survives every single exclusion (Section 4.3), so this pre-treatment fluctuation is not one municipality's pre-trend masquerading as a group pattern.

The deforestation null partly reflects sample composition. When Paragominas (PA), home to one of the largest bauxite operations in the treated group, is included, bauxite

Table 3: Bauxite: Local Effects

Outcome	Simple	Group	Obs.
Homicides (log rate/100k)	1.098	0.707	8,623
	(1.367)	(1.138)	
Employment rate (log)	-0.189^{*}	-0.200	8,560
	(0.099)	(0.194)	
Employment (log)	-0.114	-0.100	8,560
	(0.115)	(0.103)	
Wage (log, minimum wages)	-0.002	-0.031	8,560
	(0.112)	(0.081)	
GDP per capita (log, pop-wt)	$\mathbf{0.345}^{***}$	$\mathbf{0.366}^{***}$	**5,753**
	(0.107)	(0.033)	
GDP growth (log-diff, pop-wt)	-0.089	-0.089^{*}	5,479
	(0.097)	(0.046)	
Deforestation (log flow, PRODES)	-0.090	-0.115	6,302
	(0.216)	(0.094)	

Group-time ATT from Callaway & Sant'Anna (2021); inference is based on the group column. Treatment: cumulative CFEM $\geq$ R\$100k (Section 2.1); $G_i = \min(\mathit{first\_ral}_i, \mathit{first\_cfem}_i)$ (`cs_cfem_floor100k.Rda`). * $p < 0.1$, *** $p < 0.01$. Bold row is the paper's headline finding; see the text for a pre-trend caveat specific to this specification. Manufacturing employment is omitted for failing the pre-trend check (Section 4). "Obs." is defined as in Table 2.

appears to reduce deforestation by roughly one third. That effect, however, is carried entirely by a single municipality, which alone accounts for over half the treated group's area weight.[6]

4.3 Cohort-Size Robustness

The iron and bauxite estimates rest on small treated groups, so we probe them directly for single-unit influence. We re-estimated each specification once per treated municipality with that municipality excluded, for the three coefficients exposed to this concern: GDP per capita for both minerals and iron's manufacturing employment. No exclusion overturns any of them. Iron's GDP per capita stays between $+0.162$ and $+0.285$ across its 21 exclusions, significant at 5% in all but one – dropping Caetité (BA) moves it to $p < 0.1$ with the point estimate essentially unchanged, a loss of precision rather than of effect. Bauxite's GDP per capita ranges from $+0.286$ to $+0.429$ across its 7 exclusions, always $p < 0.01$. Iron's manufacturing employment ranges from $+0.377$ to $+0.648$ across its 23 exclusions, always $p < 0.01$.

[6] We examine the Paragominas case in detail in Section 5.1.

A complementary check addresses cohort thinness rather than single units: we re-estimated GDP per capita after collapsing G into four-year windows, which reduces the number of effectively singleton cohorts without addressing left-censoring. Table 4 reports two window placements, offset by two years. GDP per capita remains positive and significant under the group aggregation in every version, for both minerals. The point estimate, however, is sensitive to window phase where few usable windows remain: iron moves between +0.153 and +0.263 around its baseline of +0.227, while bauxite is stable under the 1945 anchor (+0.333) but nearly doubles under the 1947 anchor (+0.702).

Table 4: Robustness: GDP per Capita Under Four-Year Cohort Windows

Mineral	Specification	Simple	Group
Iron	Baseline (exact year, 14 usable cohorts)	0.273^{**}	0.227^{***}
Iron	4-year windows, anchor 1945 (5 usable windows)	0.185^{**}	0.153^{***}
Iron	4-year windows, anchor 1947 (6 usable windows)	0.312^{***}	0.263^{***}
Bauxite	Baseline (exact year, 5 usable cohorts)	0.345^{***}	0.366^{***}
Bauxite	4-year windows, anchor 1945 (3 usable windows)	0.312^{***}	0.333^{***}
Bauxite	4-year windows, anchor 1947 (3 usable windows)	0.666^{***}	0.702^{***}

Log GDP per capita, population-weighted, floor sample: cumulative CFEM $\geq$ R\$100k, $G_i = \min(first\_ral_i, first\_cfem_i)$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. "Anchor" is the starting year of the four-year windows. "Usable" counts cohorts (windows) beginning after the GDP panel's first year; binning shrinks this count because windows that straddle the panel's start are dropped by the estimator.

The leave-one-out exercises establish that no single municipality drives the result. A separate concern is whether standard errors can be trusted when the number of treated clusters is small (Cameron & Miller, 2015). We address this with a randomization-inference test. The logic is simple: if mining had no effect, then assigning treatment to different municipalities should produce estimates similar to the ones we obtained. We therefore simulated alternative lotteries for each mineral, randomly reassigning the observed production start years across the pool while keeping the number of treated units and start years fixed, and re-estimated the GDP-per-capita specification. The resulting 300 placebo estimates trace out the distribution of effects one would expect under a true null. Our estimates sit far in the right tail of that distribution (Figure A2, Appendix). Only 0.3% of iron placebos and 1.3% of bauxite placebos are as large as the actual estimates (randomization p-values of 0.003 and 0.013), confirming that the results do not rest on asymptotic approximations. The bauxite placebo distribution is centered slightly above zero (mean +0.085), a mechanical artifact of its small pool. Netting that out puts the bauxite gain closer to 32% than to 44%, reinforcing the caveat that bauxite's magnitude is less settled than its sign.

4.4 *Gold*

Table 5 reports outcomes for gold. GDP per capita, GDP growth, wages, homicides, and deforestation are all null under the group aggregation. The formal employment rate rises by $+0.213$ log points ($p < 0.05$ group, $p < 0.1$ simple, $\approx$24%), with no significant pre-treatment deviation in the five years preceding treatment. Total employment moves in the same direction but is not significant ($+0.165$), which cautions against a strong reading: a rate can rise because formal jobs grow or because population declines, and the level coefficient does not yet separate the two.

Table 5: Gold: Local Effects

Outcome	Simple	Group	Obs.
Homicides (log rate/100k)	0.843	0.564	66,809
	(0.600)	(0.542)	
Employment rate (log)	0.212^{*}	0.213^{**}	66,562
	(0.112)	(0.102)	
Employment (log)	0.107	0.165	66,562
	(0.104)	(0.108)	
Wage (log, minimum wages)	−0.036	−0.003	66,562
	(0.037)	(0.018)	
GDP per capita (log, pop-wt)	0.142	0.137	44,691
	(0.114)	(0.110)	
GDP growth (log-diff, pop-wt)	0.013	0.015	42,562
	(0.020)	(0.037)	
Deforestation (log flow, PRODES)	0.140	0.188	48,990
	(0.199)	(0.182)	

Group-time ATT from Callaway & Sant'Anna (2021); inference is based on the group column. Treatment: cumulative CFEM $\geq$ R\$100k (Section 2.1); $G_i = \min(\textit{first\_ral}_i, \textit{first\_cfem}_i)$ (`cs_cfem_floor100k.Rda`). * $p < 0.1$, ** $p < 0.05$. Manufacturing employment ($+0.146^{*}$ group) is omitted for failing the pre-trend check (significant deviations at $t = -1$ and $t = -2$); the employment-rate coefficient shows no significant deviation in the five years preceding treatment. "Obs." is defined as in Table 2.

We do not read the nulls on GDP, wages, crime, and deforestation as evidence that gold mining has no local footprint. Rather, we argue that CFEM is a poor treatment signal specifically for gold, because a substantial share of Brazilian gold production occurs through informal or semi-formal garimpo. If informal gold mining carries the social and environmental footprint documented elsewhere in the literature (Siqueira-Gay & Sánchez, 2021; Von der Goltz & Barnwal, 2019), our design would fail to detect it. We also examined the effects of including small-scale legal production by removing the materiality floor, which had excluded 89 gold-producing municipalities whose entire CFEM history consisted of trace payments associated with garimpo activity. When the treatment group is expanded to include these municipalities, the estimated effect on the employment rate is *attenuated, while the effect on GDP per capita remains statistically indistinguishable from zero.*

A minimum-detectable-effect (MDE) calculation can clarify whether a non-significant coefficient reflects a true absence of effect or simply a lack of statistical power to detect one. For each gold null, with estimated standard error $\widehat{\mathrm{SE}}$, we calculate the smallest true effect the design would detect with 80% power at the 5% level: $\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$. A null is informative only when this threshold falls below effects of economic interest. The gold nulls differ sharply in this respect. The wage null is tight, since the design would have detected a wage effect as small as 5% (MDE $= 0.051$). Thus,"no wage response" seems a precise finding rather than an absence of power. The GDP-growth null is similarly informative (MDE $\approx$11%). The GDP-per-capita null is only partly informative: its MDE is 0.31 log points ($\approx$36%), which exceeds the iron GDP effect ($+0.227$) but not the bauxite one ($+0.366$). Equivalently, the gold design had 91% power to detect a bauxite-sized boom but only 54% to detect an iron-sized one, so the gold GDP null rules out a large, bauxite-scale effect while leaving a gain as modest as iron's genuinely undetermined. The deforestation and homicide nulls carry the least weight: their MDEs are roughly 66% and 360% respectively, so the formal-CFEM design simply lacks the power to see the environmental and violence footprints that the literature associates with gold.

5 Robustness Checks

5.1 A Single Municipality Behind the Bauxite Deforestation Result

Under an alternative treatment definition, in which any positive CFEM payment by a titled municipality qualifies it for the treatment group and treatment timing is determined solely by the first royalty payment, bauxite mining appears to reduce deforestation by approximately one-third (-0.441, $p < 0.01$). This estimate is driven by a single case, not an average treatment effect: Paragominas (PA), host to one of the world's largest bauxite operations (first CFEM payment in 2007), spans roughly 19,300 km^2, accounting for 51.5% of the treated group's total area weight in the area-weighted deforestation specification and more than ten times the area of the second-largest treated municipality. A leave-one-out check confirms that dropping Paragominas moves the estimate from -0.44 to essentially exactly zero in both aggregations. Dropping any other treated municipality leaves the estimate essentially unchanged, and simply removing the area weights also yields a null.

Rather than discard the observation, we treat it as a single-municipality case study. Paragominas, for decades an emblem of predatory frontier development, entered the federal blacklist of the Amazon's worst deforesters in 2008 and responded by creating the *Município Verde* pact, an action plan that included environmental campaigns, environmental education projects, licensing for rural producers, deforestation monitoring, and other initiatives. The municipality subsequently became the first municipality removed from the blacklist and the template for Pará's Green Municipalities program. Hydro's bauxite mine entered commercial production in 2007, essentially simultaneously, and deforestation then fell steeply. Our design cannot separate the mine from the program it coincided with. What the case does establish, against evidence that Amazonian mining typically increases deforestation well beyond the lease footprint (Sonter et al., 2017), is that a very large, formal, royalty-paying operation proved fully compatible with one of the Amazon's best-known local deforestation turnarounds.

5.2 Does the Binary Treatment Mask Production Intensity?

Our treatment is binary, and although the materiality floor removes economically negligible producers, the remaining treated group still spans several orders of magnitude. We address this heterogeneity by splitting each mineral's treated group into quartiles of CFEM per capita. This exercise uses the CFEM-timed sample of Section 5.1, where CFEM per capita is observed for every treated unit. Each quartile is re-estimated against the common control group of never-producers, with the other quartiles excluded rather than recoded as controls. For iron, the growth effect is not concentrated among high-intensity producers. It is significant in the bottom, third, and top quartiles, with no monotonic gradient. For bauxite, the deforestation effect is concentrated entirely in the top quartile, which contains Paragominas. Gold, by contrast, is not uniformly null once split. The bottom quartile shows a robust decline in the employment rate, the third an increase in GDP growth, and the top an increase in the employment rate. This non-monotonic pattern complicates a pure informality reading of Section 4.4, and we flag it for formal testing before drawing conclusions.

5.3 International Prices and Cohort Heterogeneity

Treatment timing could be correlated with commodity price cycles: municipalities may be more likely to begin production during a price boom, in which case part of what we attribute to mining could reflect the price level prevailing at entry. We classify each iron cohort's first-CFEM year (CFEM-timed sample, as above) as "boom" or "bust" according to whether the international iron ore price (World Bank Pink Sheet) was above or below its long-run log-linear trend, and re-estimate separately for the 24 boom and 27 bust cohorts, each against the common control group. The GDP growth effect is virtually identical across regimes (+0.057 in both, significant in each), which rules out the price-timing explanation.

6 Final Remarks

This paper examines the local effects of legal mineral production in Brazil. We propose a mining lottery comparing municipalities that found economically viable resources and brought them into commercial production with municipalities that prospected but did not. Our results show that producing iron and bauxite municipalities settle onto a permanently higher level of local GDP per capita, roughly 25% and 44%, respectively. For iron, the gain comes with modestly higher wages, more manufacturing employment, and less deforestation, though with no change in overall employment or crime. For bauxite, nothing else moves. Gold delivers no detectable GDP gain at the pooled level, consistent with a large informal sector that a formal royalty record cannot capture. The iron and bauxite gains survive dropping any single treated municipality, collapsing cohorts into four-year windows, and a randomization-inference test that rejects the null.

The GDP gains, while real, follow a pattern that should reduce enthusiasm about mining as a development engine. Large mining operations arrive in small municipalities and their output enters the local GDP accounts. What stays behind is modest. In iron, a small industrial nucleus of better-paid workers in processing, maintenance, and transport. In bauxite, not even that. Overall employment does not rise in either mineral, and post-treatment growth rates are flat or marginally negative, consistent with a one-time level shift rather than a permanently faster growth path. Mining makes municipalities richer in the accounts without making them more economically dynamic. The first-order policy question is not whether to permit mining, but how to make its fiscal windfall permeate the local labor market.

A second finding is that social and environmental costs that dominate the evidence on illegal and artisanal mining do not appear in our data for the legal, royalty-paying sector. Homicide rates are statistically indistinguishable from the control group in all three minerals. Deforestation, if anything, falls in iron. Legal mining and environmental quality are at least compatible, and may be complementary where formal operations displace more destructive land uses. This distinction matters for policy: treating legal and illegal mining as a single category conflates two very different settings with very different local consequences.

A third finding concerns heterogeneity across minerals. Iron, bauxite, and gold leave very different local footprints. A single-commodity design could not have recovered this result. Iron transmits partially to wages and manufacturing employment while bauxite does not. Gold, where informal production dominates, is largely invisible to the formal royalty record. Policies designed around one mineral's enclave pattern may be poorly suited to another. This matters because the public debate on mining typically treats the sector as a single block. The evidence suggests otherwise: what is being mined, and how, shapes local consequences as much as the fact of mining itself.

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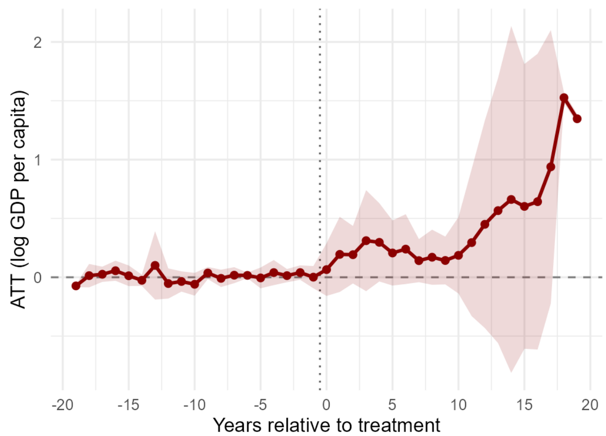
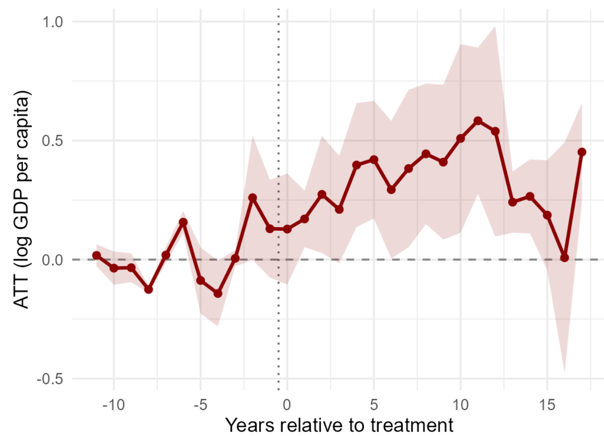
Appendix

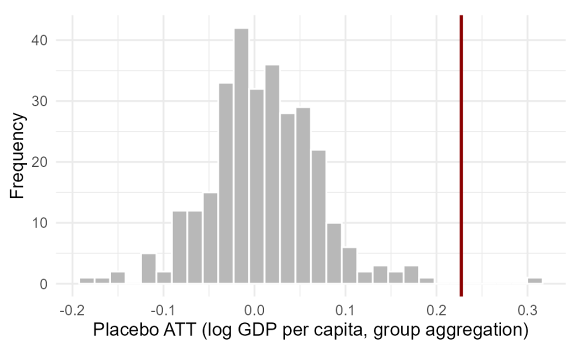


(a) Iron

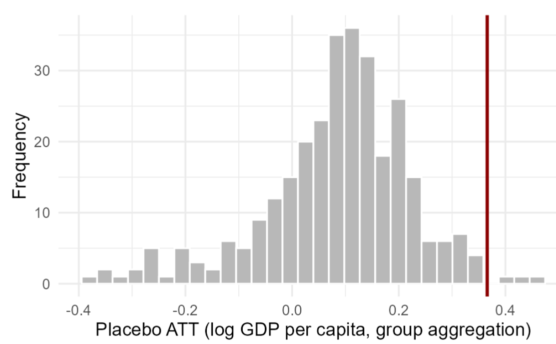


(b) Bauxite

Figure A1: Event Studies: GDP per Capita Under RAL Timing (Group-Time ATT). Outcome: log GDP per capita, population-weighted. Shaded areas are 95% simultaneous confidence bands (multiplier bootstrap).

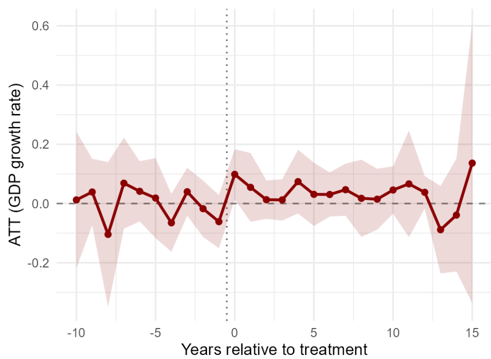


(a) Iron

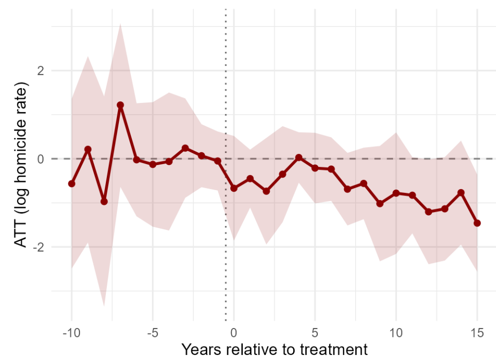


(b) Bauxite

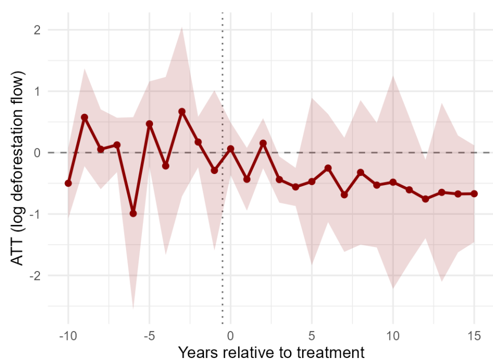
Figure A2: Randomization Inference: Placebo Distributions for GDP per Capita. Histograms of 300 placebo estimates per mineral, obtained by randomly reassigning the observed production start years within each lottery pool and re-estimating the group-aggregation ATT (Section 4.3). The vertical line marks the actual estimate (+0.227 for iron, +0.366 for bauxite); two-sided randomization p-values are 0.003 and 0.013. The bauxite placebo distribution is centered slightly above zero (mean +0.085), the mechanical tilt discussed in the text.



(a) Iron: GDP growth



(b) Iron: homicides



(c) Bauxite: deforestation

Figure A3: Event Studies: Results That Appear Only Under Registry-Only Timing (Group-Time ATT). Outcomes: log-difference GDP growth and log homicide rate per 100,000, both population-weighted; log deforestation flow (PRODES), area-weighted. Shaded areas are 95% simultaneous confidence bands (multiplier bootstrap), estimated over the full event window; the display is truncated to $[-10, +15]$ because the longest pre-treatment leads rest on very few treated units.